

## Practical (2b)

```
(kali@kali)-[~]
$ nmap -sn 10.0.2.15/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:24 EDT
Nmap scan report for 10.0.2.2
Host is up (0.00012s latency).
MAC Address: 52:55:0A:00:02:02 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.00030s latency).
MAC Address: 52:55:0A:00:02:03 (Unknown)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 13.66 seconds
```

### 1) TCP Connect Scan

```
(kali@kali)-[~]
$ sudo nmap -sT --top-ports 20 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:49 EDT
Nmap scan report for 10.0.2.3
Host is up (0.0071s latency).

PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
25/tcp    filtered smtp
53/tcp    open  domain
80/tcp    filtered http
110/tcp   filtered pop3
111/tcp   filtered rpcbind
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   filtered imap
443/tcp   filtered https
445/tcp   filtered microsoft-ds
993/tcp   filtered imaps
995/tcp   filtered pop3s
1723/tcp  filtered pptp
3306/tcp  filtered mysql
3389/tcp  filtered ms-wbt-server
5900/tcp  filtered vnc
8080/tcp  filtered http-proxy
MAC Address: 52:55:0A:00:02:03 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 5.73 seconds
```

```
(kali@kali)-[~]
$ nmap -sT -v 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:39 EDT
Initiating ARP Ping Scan at 05:39
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 05:39, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 05:39
Completed Parallel DNS resolution of 1 host. at 05:39, 5.52s elapsed
Initiating Connect Scan at 05:39
Scanning 10.0.2.3 [1000 ports]
Discovered open port 53/tcp on 10.0.2.3
Completed Connect Scan at 05:39, 0.05s elapsed (1000 total ports)
Nmap scan report for 10.0.2.3
Host is up (0.00044s latency).
Not shown: 999 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 52:55:0A:00:02:03 (Unknown)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.69 seconds
Raw packets sent: 1 (28B) | Rcvd: 1 (28B)
```

## 2) TCP Syn Scan

```
(kali㉿kali)-[~]  
$ nmap -sS -v 10.0.2.3  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:40 EDT  
Initiating ARP Ping Scan at 05:40  
Scanning 10.0.2.3 [1 port]  
Completed ARP Ping Scan at 05:40, 0.04s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 05:40  
Completed Parallel DNS resolution of 1 host. at 05:40, 5.51s elapsed  
Initiating SYN Stealth Scan at 05:40  
Scanning 10.0.2.3 [1000 ports]  
Discovered open port 53/tcp on 10.0.2.3  
Completed SYN Stealth Scan at 05:40, 0.06s elapsed (1000 total ports)  
Nmap scan report for 10.0.2.3  
Host is up (0.00011s latency).  
Not shown: 999 filtered tcp ports (net-unreach)  
PORT      STATE SERVICE  
53/tcp    open  domain  
MAC Address: 52:55:0A:00:02:03 (Unknown)  
  
Read data files from: /usr/share/nmap  
Nmap done: 1 IP address (1 host up) scanned in 5.73 seconds  
Raw packets sent: 1001 (44.028KB) | Rcvd: 1001 (72.000KB)
```

## 3) Xmas Scan

```
(kali㉿kali)-[~]  
$ nmap -sX -v 10.0.2.3  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:40 EDT  
Initiating ARP Ping Scan at 05:40  
Scanning 10.0.2.3 [1 port]  
Completed ARP Ping Scan at 05:40, 0.04s elapsed (1 total hosts)  
Initiating Parallel DNS resolution of 1 host. at 05:40  
Completed Parallel DNS resolution of 1 host. at 05:41, 5.62s elapsed  
Initiating XMAS Scan at 05:41  
Scanning 10.0.2.3 [1000 ports]  
Completed XMAS Scan at 05:41, 0.07s elapsed (1000 total ports)  
Nmap scan report for 10.0.2.3  
Host is up (0.00010s latency).  
All 1000 scanned ports on 10.0.2.3 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 52:55:0A:00:02:03 (Unknown)  
  
Read data files from: /usr/share/nmap  
Nmap done: 1 IP address (1 host up) scanned in 5.83 seconds  
Raw packets sent: 1001 (40.028KB) | Rcvd: 1001 (40.028KB)
```

#### 4) Null Scan

```
(kali@kali)-[~]
$ nmap -sN -v 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:41 EDT
Initiating ARP Ping Scan at 05:41
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 05:41, 0.10s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 05:41
Completed Parallel DNS resolution of 1 host. at 05:41, 5.55s elapsed
Initiating NULL Scan at 05:41
Scanning 10.0.2.3 [1000 ports]
Completed NULL Scan at 05:41, 0.07s elapsed (1000 total ports)
Nmap scan report for 10.0.2.3
Host is up (0.00010s latency).
All 1000 scanned ports on 10.0.2.3 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 52:55:0A:00:02:03 (Unknown)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds
Raw packets sent: 1001 (40.028KB) | Rcvd: 1001 (40.028KB)
```

```
(kali@kali)-[~]
$ sudo nmap -sN -v --top-ports 20 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:50 EDT
Initiating ARP Ping Scan at 05:50
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 05:50, 0.05s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 05:50
Completed Parallel DNS resolution of 1 host. at 05:50, 5.51s elapsed
Initiating NULL Scan at 05:50
Scanning 10.0.2.3 [20 ports]
Completed NULL Scan at 05:50, 0.02s elapsed (20 total ports)
Nmap scan report for 10.0.2.3
Host is up (0.00021s latency).

PORT      STATE SERVICE
21/tcp    closed ftp
22/tcp    closed ssh
23/tcp    closed telnet
25/tcp    closed smtp
53/tcp    closed domain
80/tcp    closed http
110/tcp   closed pop3
111/tcp   closed rpcbind
135/tcp   closed msrpc
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
445/tcp   closed microsoft-ds
993/tcp   closed imaps
995/tcp   closed pop3s
1723/tcp  closed pptp
3306/tcp  closed mysql
3389/tcp  closed ms-wbt-server
5900/tcp  closed vnc
8080/tcp  closed http-proxy
MAC Address: 52:55:0A:00:02:03 (Unknown)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.69 seconds
Raw packets sent: 21 (828B) | Rcvd: 21 (828B)
```

## 5) UDP Scan

```
(kali㉿kali)-[~]
$ nmap -sU -v 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:41 EDT
Initiating ARP Ping Scan at 05:41
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 05:41, 0.03s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 05:41
Completed Parallel DNS resolution of 1 host. at 05:41, 5.60s elapsed
Initiating UDP Scan at 05:41
Scanning 10.0.2.3 [1000 ports]
Discovered open port 53/udp on 10.0.2.3
Completed UDP Scan at 05:41, 0.12s elapsed (1000 total ports)
Nmap scan report for 10.0.2.3
Host is up (0.00016s latency).
Not shown: 999 filtered udp ports (net-unreach)
PORT      STATE SERVICE
53/udp    open  domain
MAC Address: 52:55:0A:00:02:03 (Unknown)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.86 seconds
Raw packets sent: 1046 (49.378KB) | Rcvd: 1046 (78.504KB)
```

```
(kali㉿kali)-[~]
$ sudo nmap -sU -v --top-ports 20 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-24 05:50 EDT
Initiating ARP Ping Scan at 05:50
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 05:50, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 05:50
Completed Parallel DNS resolution of 1 host. at 05:50, 5.59s elapsed
Initiating UDP Scan at 05:50
Scanning 10.0.2.3 [20 ports]
Discovered open port 53/udp on 10.0.2.3
Completed UDP Scan at 05:50, 0.08s elapsed (20 total ports)
Nmap scan report for 10.0.2.3
Host is up (0.0026s latency).

PORT      STATE SERVICE
53/udp    open  domain
67/udp    filtered dhcps
68/udp    filtered dhcpc
69/udp    filtered tftp
123/udp   filtered ntp
135/udp   filtered msrpc
137/udp   filtered netbios-ns
138/udp   filtered netbios-dgm
139/udp   filtered netbios-ssn
161/udp   filtered snmp
162/udp   filtered snmptrap
445/udp   filtered microsoft-ds
500/udp   filtered isakmp
514/udp   filtered syslog
520/udp   filtered route
631/udp   filtered ipp
1434/udp  filtered ms-sql-m
1900/udp  filtered upnp
4500/udp  filtered nat-t-ike
49152/udp filtered unknown
MAC Address: 52:55:0A:00:02:03 (Unknown)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds
Raw packets sent: 33 (2.378KB) | Rcvd: 31 (2.994KB)
```

